TRACK: RESTORATION

SeedStick

One push.
One seed ball.
Every spot on record.

A hand-held planting tool for land restoration. It checks the soil, sets each seed ball at the same depth, and counts every one, so planting goes faster without losing care.

FIG. 1

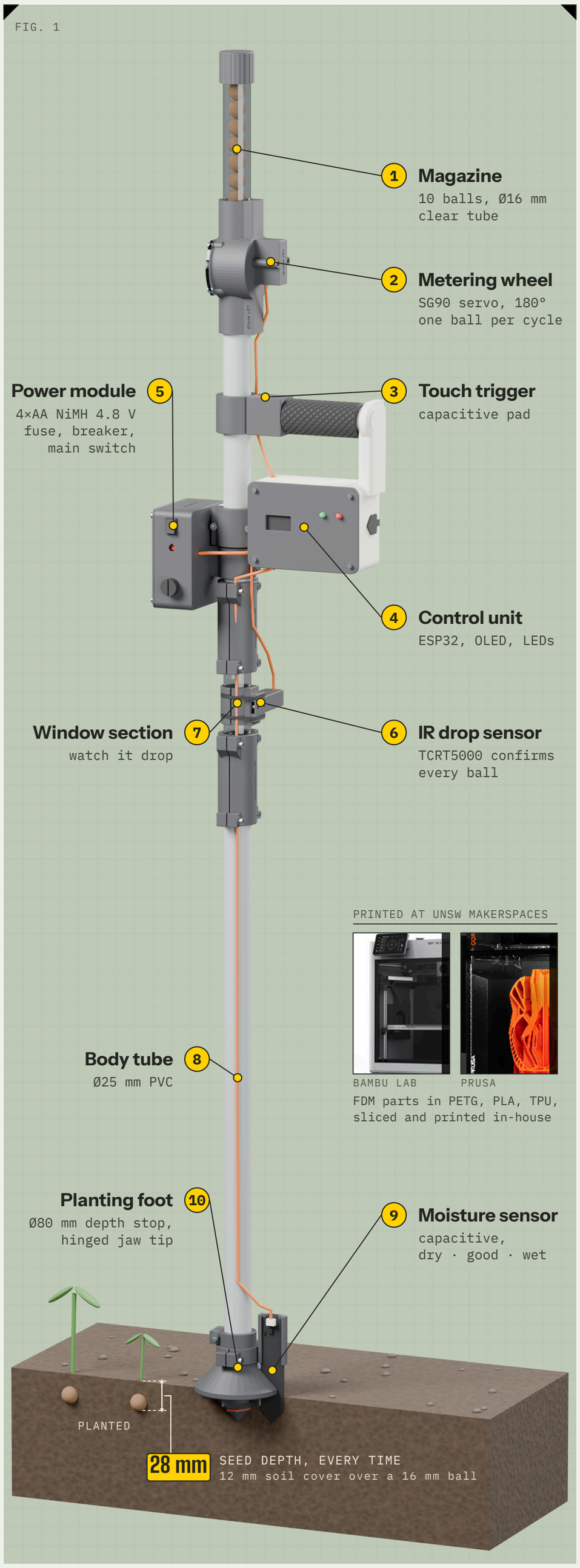


FIG. 1 — SEEDSTICK PROTOTYPE IN THE SOIL. BLENDER RENDER.

01 Restoration still moves at hand speed

1.2 bn ha
of degraded land pledged for restoration by 2030

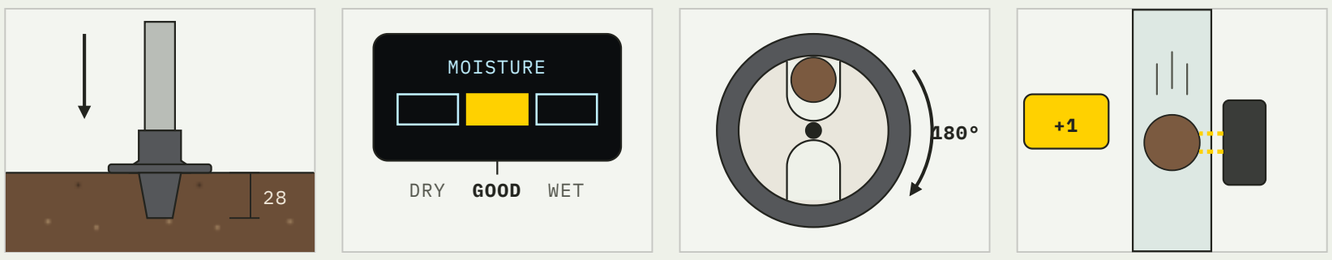
10.4 %
of that area is actually under restoration today

US$1.6 tn
needed this decade, because the work is slow and manual

Seedlings are still placed by hand, a few hundred per person per day. Seeds sown over large areas often germinate poorly.

FIGURES: BUILD-A-THON 2026 BRIEF, PROBLEM 3: RESTORATION

02 Push. Read. Release. Confirm.



1 Push in
The depth stop sets every seed ball exactly 28 mm deep.

2 Read the soil
The moisture sensor shows dry, good or wet before you commit.

3 Release one
A touch turns the metering wheel 180°. Exactly one ball drops.

4 Confirm & log
An IR sensor sees it pass. Count, time and soil band are saved.

03 What it changes

Less labour — Plant standing up: push, touch, move on. No bending, no digging, no counting.

Better starts — Every ball at exactly 28 mm, and a soil check at every spot before it goes in.

Proof of work — Every planting point is counted and logged, readable over the ESP32's own Wi-Fi.

04 Safe by design

Low voltage
4 × AA NiMH, 4.8 V. Far below the 12 V limit.

Double protection
Glass fuse plus a resettable circuit breaker.

One main switch
A single rocker switch cuts all power.

Guarded mechanism
Wheel behind a clear cover, servo under a guard.

Bolted together
Every module is bolted on with heat-set inserts.

Sealed battery bay
Screwed lid: the cells only come out with a tool.

Try it at our table

Push it in, touch the pad, and watch exactly one seed ball drop.

AUSTRALIA × GERMANY

The best mix of engineering. *Trust the engineers.*